



Syllabus

This is a hybrid, flipped, just-in-time class

This class is a *hybrid, flipped, just-in-time* class. **Hybrid** means we will have a Zoom session associated with the class. The class is hybrid for the sake of equity. We do not want students unable to attend due to external factors outside their control. However, you are expected to attend class in person if at all possible. We have evidence that attending class online may not support your learning as well as if you were in person. Therefore, you must request to attend a class online to receive the password for that day on the [forms page](#). For more details, see the [policies page](#). **Flipped** means you learn the easier-to-grasp material before class and have already made an initial attempt at understanding it, including some simple comprehension quizzes. Then, we spend class meaningfully engaging with the material and learning the harder-to-grasp material. **Just-in-time** means we use the data from the quizzes to drive what happens in class. Based on the quiz results, class focuses on what everyone least understood from the material and allows everyone to apply what they learned using example questions. Solutions to the examples are released after class.

Modules

This course is divided into modules that are typically 1 week long. Each module has multiple components, so you engage with the material multiple times to enhance your learning. All due dates are at 11:59 PM unless otherwise noted.

- **Prepare [Due: Typically Monday].** Will typically contain videos and/or readings to introduce new content, as well as quizzes (usually one per video/reading) to ensure comprehension. The quizzes will be on Canvas and auto-graded.
 - You can take the quizzes **up to three times** to correct initial misunderstandings.
 - A score of **80%** total for the quizzes in that specific Prepare (e.g. $\text{sum}(\text{points_earned_on_all_quizzes})/\text{sum}(\text{points_possible_on_all_quizzes})$) will result in full credit for that Prepare. Otherwise, the percentage grade is based on 80% of the possible points (so a total of 72 points out of 100 results in 90%).
 - These are due 2 days before class, so we have enough time to tailor class based on everyone's understanding in the Prepares.
 - There is a 24-hour late window (typically the entire Tuesday).
 - You are allowed **one** extension request on **one** Prepare, no questions asked. Request this extension on the [forms page](#). For additional extensions, you must email the course email address.
- **Class Participation [Due: day of class].** During class, we will have active learning activities, such as peer instructions and think-pair-shares, to help you develop your understanding as we go through the content. They are created based on the class's performance on the Prepares, so they focus on what everyone needs the most help in learning. Peer instructions are questions that you answer individually, then discuss with your peers, and then answer the question again to help cement everyone's understanding. Think-pair-shares involve an open-ended question that you think about alone first, then discuss with others, and some groups will share with the class. Peer instructions involve two forms, while think-pair-shares involve a single form. These active learning activities will be available **ONLY** while they are being used during class.
- **Worked Examples.** During class, we will also work on notebooks together. They do not factor into your grades and are not available for every module. When they are available, they come with an attached autograder that you can run locally to see if you have solved the problems correctly.
- **Homework [Due: Typically Sunday].** Will typically consist of a set of coding/data analysis or statistics problems to complete in a Jupyter

notebook and submit on Gradescope. You can start these right after finishing the Prepare and before class. We may work through several of these problems in class together, especially if there are no worked examples. You may submit as many times as you want to Gradescope. A score of **90%** on a Homework will result in full credit. The formula is as follows:

$$\circ \quad \min(1, \text{score}/(\text{points_possible} * 0.9))$$

Homework Late Days

Homework will typically be due on Sunday. If you want to submit late, you must use late-day “tokens.” You have 9 late days for the entire semester. To use your late days, you need to fill out the late day form on the [forms page](#) within 24 hours of the due date (before or after), so we are aware to expect a submission and a rough idea of how many days you plan to use. To submit, submit your homework like normal during Gradescope’s (usually) 1-week late period. The total number of late days you use will be based on the calendar day in Duke’s time zone when your active submission was submitted, not what was stated on the form. You do not need to submit the form more than once per assignment.

For example, if the assignment was due on Sunday and your active submission was submitted on Tuesday, you used 2 late days for that assignment. Assuming this is the first time you used late days, you will have 7 late days left.

In the Canvas grade book, you will find a column telling you how many late days you have used. If you run out of late days, you will need to contact the professor through the class email (see [course info page](#)) to discuss getting more. The “denominator” of the column will be the number of default tokens anyone has. If you receive more, we cannot change that number individually for you. You will need to keep track of how many tokens you can spend yourself.

Final Submissions Must be a Fresh Restart and Run All

The notebook for your final homework submission must have a [fresh restart and run all](#). This is important because a common way to detect bugs that the autograder might find is to first restart the kernel and run everything. Moreover, it is the equivalent of ensuring that you are submitting a polished notebook.

If your active homework submission is not a fresh restart and run all, you will **lose 3% of the possible points for that homework**. Note a score of **90%** on a Homework will result in full credit. Therefore, as long as you are doing well, this will not harm you. However, to convey the importance of this, we are making it part of your grade.

This penalty does not apply to exams. However, you are strongly encouraged to do this since a fresh start is a common way to detect bugs in your code.

Class Engagement Points

Student engagement in the education research field is defined as a student's active, emotional, and cognitive participation in the learning process. This class cannot grade your emotional participation, but we can define actions you can take to demonstrate active and cognitive participation. Engaging in your learning is vital for learning course material such that you retain it for the long term. Therefore, we have class engagement points as part of your grade. By the end of the semester, you must earn at least 60 points to get full credit. You can earn points in the following ways:

- For each class day:
 - 1 point – Fill out at least 1 form/activity during class.
 - 1 point – Fill out at least 75% of the forms/activities used in class (peer instructions involve 2 forms and think-pair-shares involve 1 form). We will (mostly) round down the number of forms/activities that need to be done each day. So, if there is only 1 form/activity and you fill it out, you will earn this point. If there are 2, you only need to fill out 1. If there are 3, you need to fill out 2. Etc.
 - 1 point – Fill out N-1 of the N available forms/activities.
- 1 point – Fill out a survey. There will be at least 3 surveys.
- 2 points – Each exam will have a study quiz in Canvas. Each question is worth a fraction of a point, such that if you earn a perfect score on the study quiz, you will earn 2 points (the best score is in the grade book). There are no limits on the number of attempts on these quizzes. There are 2 exams.
- 1 point – Speak up in class by asking/answering a question or sharing what your group discussed with the rest of the class.
- 1 point – Do a module's find data science in the real world challenge
- ? points – Reach out to the professor and propose a way to earn points!

Note that there are 10 modules and 2 days per module. Therefore, attending and actively engaging during every module's class day will earn you 60 points. Moreover, since things outside of your control can happen to cause you to miss a class, there are plenty of ways to earn points so you can still achieve the 60 points. Finally, your grade for this part of your overall grade is the $\min(\text{your_points}/60, 1)$.

Exams

There will be two exams to assess understanding of the material. These exams will have two parts: a Practicum and an in-person component.

Practicum

The Practicum focuses on your ability to produce code and interpret the results of that code. You may work solo or in pairs. The Practicum will be open for 48 hours and take 3-5 hours. The day of the Practicum will be the same day as a class period, and class will be canceled on that day to compensate for the exam.

You may use whatever resources you want for the Practicum. The focus is more on the production of code (so yes, the use of LLMs is allowed) and your understanding of what the code is doing (so even if an LLM produced the code, you are still responsible for understanding it and will be graded as such).

In-Person Exam

The in-person exam focuses on your skills to think like a data scientist and content that would be easy for an LLM to answer. They are your opportunity to show yourself and the teaching staff how much of the material you have mastered without the crutch of the Internet or tools like LLMs.

Practicum Update/In-Person Exam Retake

The goal of this course is to ensure your grade reflects what you know by the end of the course. Therefore, there are second opportunities to show your understanding and improvement compared to your first attempt.

The **Practicum Update** will happen after the Practicum. Your group will receive feedback on your Practicum and will have 3 days to respond to that

feedback to update your Practicum submission. You can then submit an updated version and a change log of what your group updated. Your overall Practicum grade will be a weighting of the original and update.

The **In-Person Exam Retakes** will happen during the Final Exam period, check DukeHub for when the final exam period is. Each retake will take up an equal part of the final exam period.

Regrades

Regrades are through Gradescope. The regrade window opens 24 hours after the assignment is returned and closes after 1 week. The 24-hour delay allows everyone time to consider their grade and consult with each other and the teaching staff before submitting a regrade. Regrade requests should explain specifically why you believe a different grade is more appropriate, not just ask for more partial credit without any reason. Regrades requests also mean that your entire submission may get regraded. Please note that grade changes (apart from clear grader errors) are rare.

Projects

Instead of a final exam, this course has an open-ended collaborative project. In groups of four or five, you will choose a research topic that can be explored through data science. You will formulate research questions, acquire data, and perform your own data processing, analysis, and modeling to answer your research questions. The projects have three stages of deliverables.

- Initial Plan – A document that includes the product of a brainstorm, reflection, and a collaboration plan.
- Proposal – A document highlighting a topic, data source(s), and research questions.
- Prototype – A document highlighting methods, preliminary results, and a reflection.
- Final Report and Video Presentation – A document that provides a complete description of the topic, research questions, methods, results, and conclusions, along with a recorded presentation involving all group members.

More details will be provided about each deliverable closer to their due dates, including a rubric for grading. Project Deliverables will be graded on a scale

where exemplary work will get 100% of the possible points and satisfactory will earn 90% of the possible points per grading criteria, in other words some kind of A. This is intended to encourage creativity to allow project groups to explore widely in terms of topics, research questions, and methods without fear of a grade penalty. We will also provide constructive written feedback on project deliverables separate from the criterion grading in order to help teams make progress.

Grades

The final course grade, as a percentage, will be calculated as the following weighted average:

- Prepare 15%
- Class Engagement Points 5%
- Homework 25%
- Project 10%, broken down by deliverables:
 - Initial Plan 0.5%
 - Proposal 1%
 - Prototype 2%
 - Video Presentation 3%
 - Final Report 3.5%
- Exams 45% (Practicum/In-Person Exam)
 - Exam 1: 9.5% / 16.5% (this exam covers more modules)
 - Exam 2: 7% / 12%

Final numerical grades will be converted to letter grades as follows. Letter grades of A+ are awarded only for students with a grade of a high A and exceptional course projects, as determined by the instructor.

- [90, 95) = A-, [95, 100] = A
- [80, 83) = B-, [83, 87) = B, [87, 90) = B+
- [70, 73) = C-, [73, 77) = C, [77, 80) = C+
- [60, 63) = D-, [63, 67) = D, [67, 70) = D+

Note: The A to A- cutoff is atypical. This ensures that to earn an A, you must earn an A on at least one exam, if not both.

The “No Project Option”

Sometimes, life gets in the way of learning. Rather than have you give up on the class, we have created the “No Project” option. Taking this option means you will not do a project, and any project grades you have will be made a 0. Since the project is worth 10% of the overall grade, this naturally caps your grade to a B+. If you take this option, we will work with you to stretch out deadlines and figure out the Practicums such that you have time to learn the content and show us that you have gained competency in the class material.

To take this option, you must meet with Prof. Stephens-Martinez. Not submitting a project milestone is grounds for moving to this option.

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